

Barrier Functions for Dynamical Systems

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Constructing Barrier Functions for Dynamical Systems

Executive summary

Barrier functions turn a safety claim into a scalar inequality that can be checked along system trajectories. For a controlled system, the usual goal is to find a function whose superlevel set is forward invariant and whose derivative can be kept nonnegative by an admissible input. This separates three tasks that are often conflated: defining the desired safe set, constructing or parameterizing a candidate barrier, and verifying that the candidate can actually be enforced by the dynamics and input limits.

Analytical construction is strongest when a trustworthy model is available. Hand-designed functions are often easiest to interpret; sum-of-squares methods automate the search for polynomial certificates; and Hamilton–Jacobi methods provide a more systematic route to maximal robust safe sets. Data-driven construction becomes useful when the model is unknown or expensive to identify, but its guarantee depends on what the data support: model uncertainty sets, coverage assumptions, Lipschitz bounds, statistical confidence, or a falsifier that searches for counterexamples.

The main practical conclusion is that “data-driven barrier function” is not one method. It ranges from robust set-membership synthesis with deterministic guarantees to neural fitting with only empirical evidence. The guarantee type should be treated as a first-class design choice.

1. Mathematical setting

Consider a continuous-time control-affine system

$$\dot{x} = f(x) + g(x)u,$$

with admissible inputs $u \in U$. A safe set is commonly represented as

$$\mathcal{C} = \{x : h(x) \geq 0\}.$$

For a relative-degree-one constraint, a standard control barrier function condition is

$$\sup_{u \in U} \left[L_{f_h}(x) + L_{g_h}(x)u \right] \geq \alpha(h(x)),$$

where α is an extended class- $\mathcal{K}$ function. Any sufficiently regular feedback law that satisfies the corresponding inequality renders $\mathcal{C}$ forward invariant under the usual boundary regularity assumptions [Ame19].

This controlled formulation grew out of classical barrier certificates for autonomous systems. Those certificates separate initial and unsafe regions and impose a flow condition that rules out trajectories reaching the unsafe set [Ran04, Pra05]. Control barrier functions add an existential input condition, so the certificate can be used not only to verify safety but also to synthesize a controller [Wie07, Ame16].

The central distinction is therefore:

Object	Main question	Typical output
Barrier certificate	Can the modeled dynamics reach the unsafe set?	A certificate of safety
Control barrier function	Is there an admissible input that preserves safety?	A safe input constraint
Barrier-based controller	Which safe input should be applied?	Often a quadratic-program solution
Viability or barrier-value function	From which states can safety be maintained?	A larger, often maximal, safe set

2. Analytical construction

2.1 Hand-designed barriers

The simplest construction encodes the safety geometry directly. Distance-to-obstacle functions, headway constraints, joint-limit margins, energy-like functions, and combinations of physical constraints are common choices. The candidate is then differentiated along the dynamics and checked against the CBF inequality.

This approach is attractive because the function has a clear interpretation and can be designed without a large optimization problem. Its weakness is that a geometrically reasonable function need not be enforceable under the available actuation. Input bounds, high relative degree, nonholonomic motion, or competing constraints can make the resulting CBF-QP infeasible even when the original safe set contains viable states [Gar23].

For relative degree greater than one, the input does not appear in the first derivative of the constraint. High-order CBFs address this by recursively defining auxiliary functions until the control appears, then imposing a final inequality on the resulting chain [Ame19, Pan24b]. This construction is useful but introduces additional gains, feasibility conditions, and tuning choices.

2.2 Polynomial search with sum-of-squares programming

When the dynamics and the sets are polynomial or semialgebraic, barrier construction can be posed as an algebraic search. A polynomial candidate is parameterized by unknown coefficients, and nonnegativity conditions are relaxed to sum-of-squares constraints. The resulting semidefinite programs can verify a candidate or search jointly over candidate and auxiliary multiplier polynomials [Ran04, Cla21].

The Positivstellensatz provides the algebraic bridge: emptiness of a set described by polynomial inequalities can be expressed through polynomial identities involving sum-of-squares multipliers. In the CBF setting, this converts the absence of a state that violates the barrier condition into an SDP-verifiable certificate [Cla21]. The method can also handle high-degree candidates and multiple constraints, although the size of the SDP grows quickly with state dimension and polynomial degree.

Synthesis is harder than verification because the candidate coefficients and the SOS multipliers appear bilinearly. Clark proposes alternating-descent heuristics, fixed-point and Lyapunov-based initializations, and specialized searches for compact or convex regions [Cla21]. These methods are useful when the

system has a tractable polynomial representation, but they do not remove the need to choose degrees, parameterizations, and a search region.

2.3 Converse and minimal constructions

A candidate barrier is not guaranteed to exist in the smooth form that a chosen algorithm assumes. Converse results clarify when safety or invariance implies the existence of a certificate. Minimal barrier functions based on scalar comparison systems can characterize invariance with weaker regularity requirements than standard smooth CBF definitions [Kon21]. More recent converse theorems give sufficient conditions for the existence of CBFs and compatible safety-and-stability certificates, while also identifying cases where a smooth or strictly compatible pair cannot exist [Mes24].

These results are important diagnostically. Failure to find a barrier may reflect a poor template or insufficient regularity, rather than actual loss of safety. Conversely, a safe set may not be representable by a single smooth function with a globally feasible CBF inequality.

2.4 Reachability-based construction

Hamilton–Jacobi reachability offers a more constructive alternative. Its value function is obtained from a differential game or viability problem and directly describes the set of states from which safety can be maintained under bounded disturbances and input constraints. Control barrier-value functions combine this constructive route with the pointwise QP implementation of CBFs [Cho21b].

The appeal is that the zero-superlevel set can recover the viability kernel rather than an arbitrary conservative subset. The price is computational: grid-based Hamilton–Jacobi methods scale poorly with dimension, and the resulting value function may be nonsmooth. They are most useful for low-dimensional systems, offline computation, or generating safe-set and barrier-function priors.

2.5 Robust analytical construction

Model uncertainty can be included directly in the barrier inequality. Robust CBFs require the derivative condition to hold for every model in an uncertainty set. For parametric uncertainty, duality can convert the worst-case constraint into a tractable QP, while online set-membership identification can shrink the uncertainty set as data arrive [Coh22].

Disturbance robustness has a related but important qualification. Under bounded nonvanishing disturbances, the original safe set may cease to be exactly invariant; instead, a relaxed set can be rendered stable, with the relaxation determined by the disturbance bound [Xu16]. This distinction should be stated explicitly rather than described simply as “robust safety.”

3. Data-driven construction

Data-driven methods differ mainly in what is inferred from data and where the formal guarantee enters. A useful taxonomy is given by surveys of learned CBFs and safe reinforcement learning [Gue24].

Data-driven route	What is learned	Typical guarantee	Representative work
Model learning with uncertainty	Unknown dynamics and an error bound	Probabilistic or robust safety	[Jag20, Coh22]

Data-driven route	What is learned	Typical guarantee	Representative work
Direct set-membership synthesis	A consistent model set, certificate, and controller	Deterministic for all models consistent with data	[Lup21, Lav21b, Nej22]
Single-trajectory polynomial synthesis	Certificate directly from persistently exciting data	Deterministic under rank and model-class assumptions	[Sam24]
Demonstration or prior refinement	A barrier compatible with safe behavior	Deterministic only with coverage and regularity conditions	[Rob20, Dai23]
Neural, Koopman, or statistical verification	A flexible barrier representation plus verification margin	Deterministic, probabilistic, or empirical depending on method	[Zin22, Tay24, Tay25, Str25b]

3.1 Learn a model, then construct a robust barrier

Gaussian-process methods first estimate unknown dynamics and then insert the posterior uncertainty into the barrier condition. Jagtap, Pappas, and Zamani assume a known input channel and an unknown drift with bounded complexity in a reproducing-kernel Hilbert space. The GP supplies a confidence bound on the model error, and the CBF condition is enforced against the worst error within that bound [Jag20]. The resulting safety statement is probabilistic, with confidence increasing as the data reduce posterior uncertainty.

This route is attractive when derivative or dynamics measurements are available and a statistical model is credible. It is less suitable when the GP prior is poorly matched, the state distribution shifts beyond the training region, or a hard worst-case guarantee is required.

A complementary approach treats the unknown parameters as belonging to a set consistent with input-output data. Robust CBF constraints are enforced for the whole set, and integral set-membership updates shrink it online without requiring noisy state-derivative measurements [Coh22]. This produces a deterministic guarantee relative to the assumed uncertainty set, but the initial set must be known to contain the true parameters.

3.2 Direct data-to-certificate synthesis

For unknown polynomial systems, data can be used without first identifying one nominal model. Luppi and colleagues construct an outer approximation of the set of polynomial models consistent with noisy data, then use a data-dependent SOS program to design an invariant set and state-feedback controller [Lup21]. This is a robust construction: the certificate is intended to hold for every model allowed by the data and noise bound.

Other methods formulate safety verification or controller synthesis as robust or scenario optimization over unknown dynamics [Lav21b, Nej22, Nej23]. Multiple local certificates can be useful when no single global polynomial template is feasible. This relaxes the search problem, but it adds a partitioning or switching design problem.

A particularly direct discrete-time approach constructs a control barrier certificate from one measured trajectory. The key requirement is a rank or persistence-of-excitation condition on the observed monomial features. When that condition holds, an SOS program produces a certificate and controller without explicitly identifying the unknown polynomial dynamics [Sam24]. The infinite-horizon claim is formal, but it rests on full-state measurements, the polynomial model class, exact data relations, and the excitation condition.

3.3 Learn from safe demonstrations

Demonstrations provide information about safe behavior but do not, by themselves, prove invariance. Robey and colleagues address this gap by learning a candidate CBF from safe state-action data while imposing constraints on the barrier value, its derivative along demonstrated actions, and a surrounding unsafe layer [Rob20]. Lipschitz bounds and sufficiently dense ϵ -nets then extend pointwise inequalities to the continuous domain.

This is a useful template for settings where a safe expert or baseline controller exists. The formal guarantee is not simply a consequence of fitting the demonstrations. It depends on coverage of the region of interest, valid Lipschitz bounds, smoothness of the candidate, and sufficient positive slack at the sampled points [Rob20].

Barrier refinement methods start from a conservative handcrafted function and use data to expand or reshape its safe set. Prior structure can reduce sample demands, while targeted sampling near high-loss or uncertain regions improves efficiency [Dai23]. The tradeoff is that refinement inherits the weaknesses of the prior and may optimize safe-set volume without finding the maximal controlled-invariant set.

3.4 Neural and Koopman parameterizations

Neural networks provide expressive candidate classes but make certification harder. The neural Koopman approach of Zinage and Bakolas jointly learns a lifted dynamical representation and a CBF, while a falsifier searches for states that violate the learned conditions. A Lipschitz-based approximation bound transfers validity from the learned lifted model to the original unknown system [Zin22]. This is a promising learner-falsifier pattern, but it assumes bounds on the unknown dynamics and the learned observables and is sensitive to discretization.

Tayal and colleagues learn a neural CBF for stochastic systems while enforcing bounds on the network, its Jacobian, and its Hessian-related terms. A finite sample set plus a covering argument yields a formal validity condition over the continuous state space [Tay24]. The method is more than empirical loss minimization, but it still assumes known stochastic dynamics and can become conservative as the required coverage resolution grows with dimension.

Two newer directions change the verification mechanism. Conformal calibration produces probabilistic guarantees with a user-selected violation rate and confidence, potentially reducing the conservatism of global Lipschitz bounds [Tay25]. ReLU or continuous piecewise-affine barriers can instead yield deterministic guarantees for Lipschitz discrete-time systems, though triangulation and known system bounds limit scalability [Str25b].

4. Comparing guarantee types

The phrase “guaranteed safe” is incomplete without specifying the quantifiers and assumptions.

Guarantee type	Quantifier	Main assumptions	Main failure mode
Exact model-based	All trajectories of the modeled system	Correct model, regularity, feasible inputs	Model mismatch or infeasible constraints
Robust set-membership	All models in a certified uncertainty set	True model lies in the set; uncertainty description is valid	Conservative uncertainty set
Probabilistic	High-probability model or state-distribution statement	Statistical model, sampling assumptions, confidence calibration	Distribution shift or incorrect prior
Finite-sample Lipschitz	All states covered by a sampling net	Known or bounded Lipschitz constants and adequate coverage	Curse of dimensionality and conservatism
Empirical	Tested trajectories or held-out samples	Representative experiments	No formal protection between or beyond samples

This table also explains why apparently similar neural CBF papers can make different claims. A training loss that is small on sampled points is not a certificate. A falsifier can strengthen the result by searching for counterexamples, but the absence of a counterexample is only as strong as the search domain and numerical soundness. A conformal method gives a statistical statement, not a worst-case deterministic one [Gue24, Tay25].

5. Practical construction workflow

A defensible barrier-function project can be organized as follows.

1. **Define the safety semantics.** Specify the safe set, unsafe set, initial set, time horizon, disturbance model, and whether safety means exact invariance, finite-horizon avoidance, or a chance constraint.
2. **Check viability before choosing a barrier.** If input bounds make the desired set nonviable, no CBF parameterization can fix the problem. Compute or approximate a viability kernel, or deliberately search for a smaller controlled-invariant subset [Cho21b].
3. **Choose the certificate class from the model structure.** Use geometric functions for simple constraints, SOS polynomials for low-dimensional polynomial systems, HJ or CBVF methods when maximal robust sets matter, and neural or piecewise-affine classes when dimension or geometry defeats low-degree polynomials.
4. **Handle relative degree and implementation.** Use high-order constructions when required. Account for disturbances, state-estimation error, sampled-data control, actuator limits, and multiple simultaneous constraints [Gar23].
5. **Separate candidate search from verification.** First optimize the candidate and controller. Then verify the full continuous-domain inequality, using SOS, interval methods, reachability, falsification, Lipschitz margins, or calibrated statistical bounds.

6. **Report the guarantee with its quantifiers.** State whether it is deterministic or probabilistic, what set of models or disturbances is covered, what data coverage is required, and whether the controller remains feasible under the stated input limits.
7. **Measure conservatism.** Compare the certified safe set with the known safe geometry, an approximate viability kernel, or a baseline barrier. A certificate can be valid and still be unusably small.

6. Recommended tutorials and surveys

Paper	Best use	Strength	Limitation
[Ame19]	Broad first introduction to CBF theory	Clear account of definitions, invariance, QP control, and applications	Mostly model-based; data-driven construction is not its focus
[Gar23]	Advanced practical CBF design	Covers disturbances, high relative degree, input limits, sampled data, output feedback, nonsmooth barriers, and conservatism	Dense and centered on a particular research program
[Gue24]	Learning CBFs and safe reinforcement learning	Useful taxonomy of demonstrations, model learning, neural methods, and guarantee types	Survey depth varies across subtopics
[Pan24b]	Short mathematical primer	Compact treatment of standard, high-order, and adaptive CBFs	Less coverage of verification, discrete time, stochastic systems, and data-driven methods
[Xia26]	Learning for feasible and safe CBF control	Specialized tutorial on HOCBF feasibility, parameter learning, and learned feasibility constraints	Not a general survey; focuses on QP feasibility and learning

For a first pass, [Ame19] is the best broad entry point. [Gar23] is the most useful follow-up for implementation issues. For the data-driven side, [Gue24] should be read alongside one or two primary methods such as [Rob20], [Jag20], [Lup21], or [Sam24]. [Xia26] is most useful when the main problem is not merely safety, but maintaining feasibility of a CBF-QP during learning or execution.

7. Open problems

Several gaps recur across both analytical and data-driven work.

- **Construction versus verification.** Verifying a proposed barrier is often easier than finding one with a large, feasible safe set.
- **Input-constrained viability.** Many elegant CBF conditions become infeasible near the boundary when actuators saturate or constraints conflict [Gar23].

- **Scalability.** SOS and grid-based reachability suffer from dimension; neural methods shift the burden to verification, coverage, and robustness bounds.
- **Distribution shift.** Statistical and learned certificates can fail outside the state-action distribution represented by the data unless uncertainty or coverage is controlled.
- **Compositionality.** Multiple obstacles and agents create feasibility and regularity problems that are not solved by simply multiplying or summing individual barriers.
- **Nonsmooth and hybrid dynamics.** Real systems often include contact, switching, sampled control, and discontinuous policies; classical smooth CBF assumptions then need careful replacement [Gar23].
- **Less conservative guarantees.** Reachability, conformal calibration, adaptive uncertainty sets, and data-efficient refinement all aim to enlarge certified safe regions without weakening the claim [Cho21b, Coh22, Dai23, Tay25].

The most reliable research direction is therefore hybrid rather than purely analytical or purely learned: use physics, geometry, or reachability to supply a structured prior; use data to reduce model uncertainty or refine the candidate; and retain an explicit verification layer that matches the intended safety claim.

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